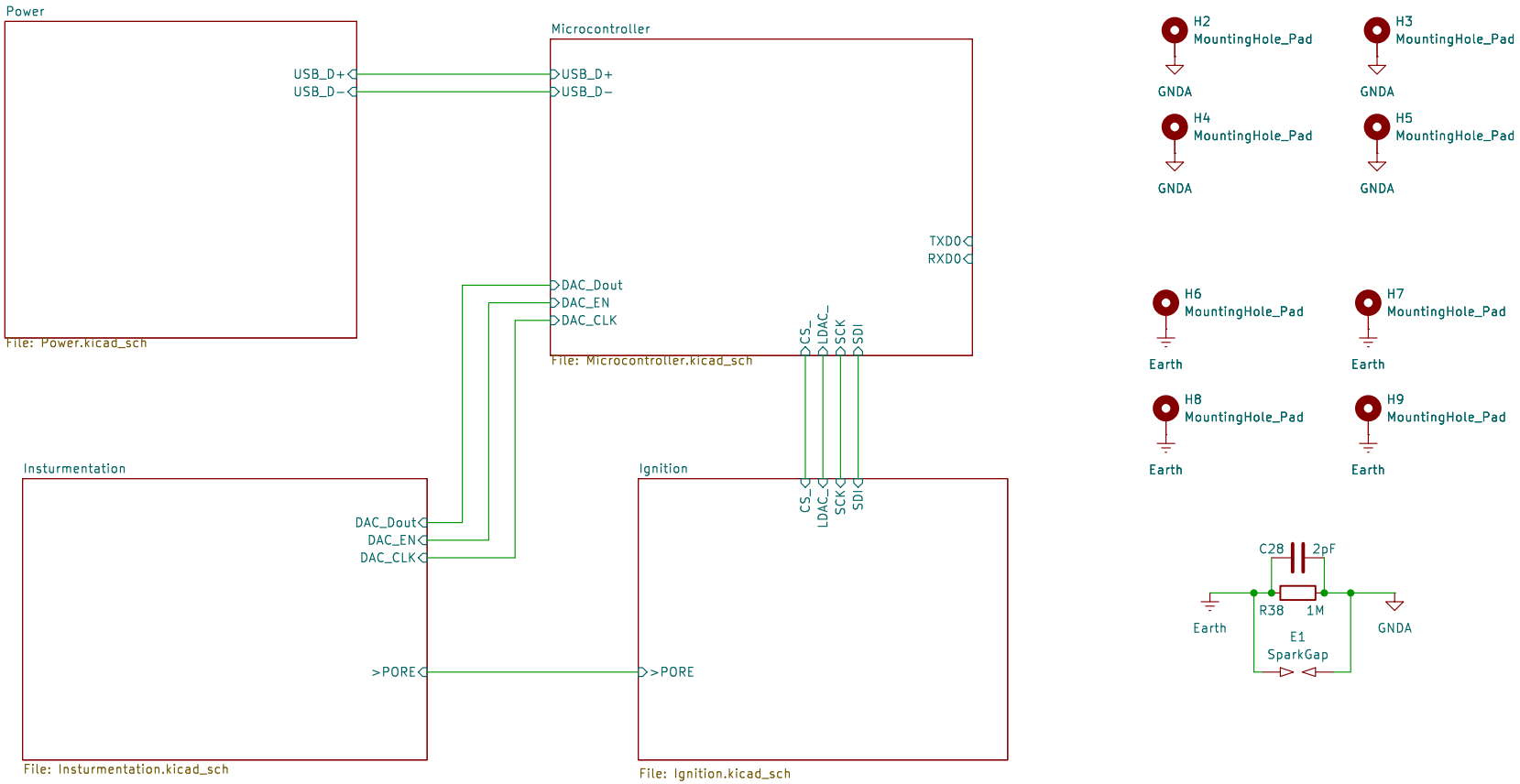
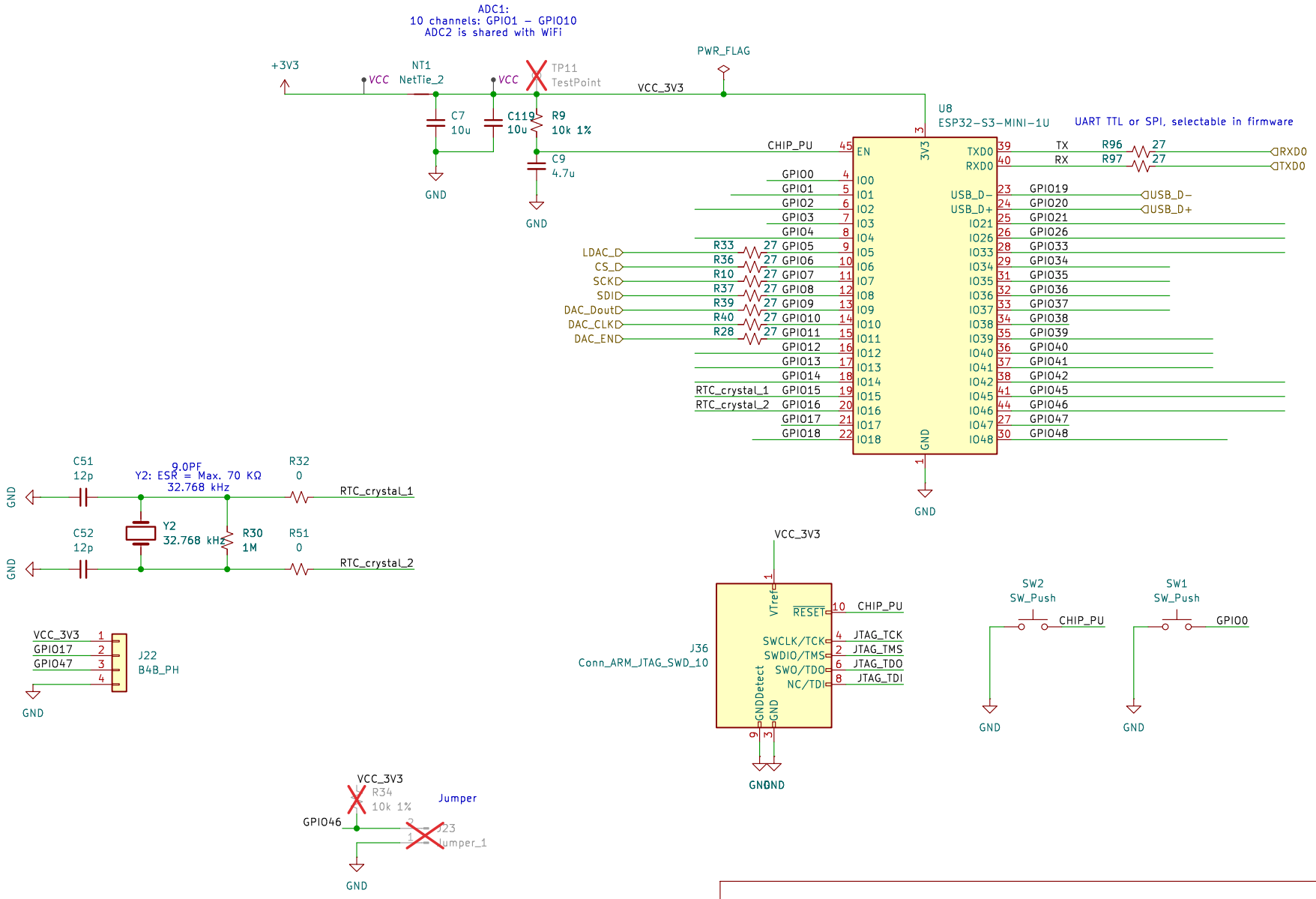
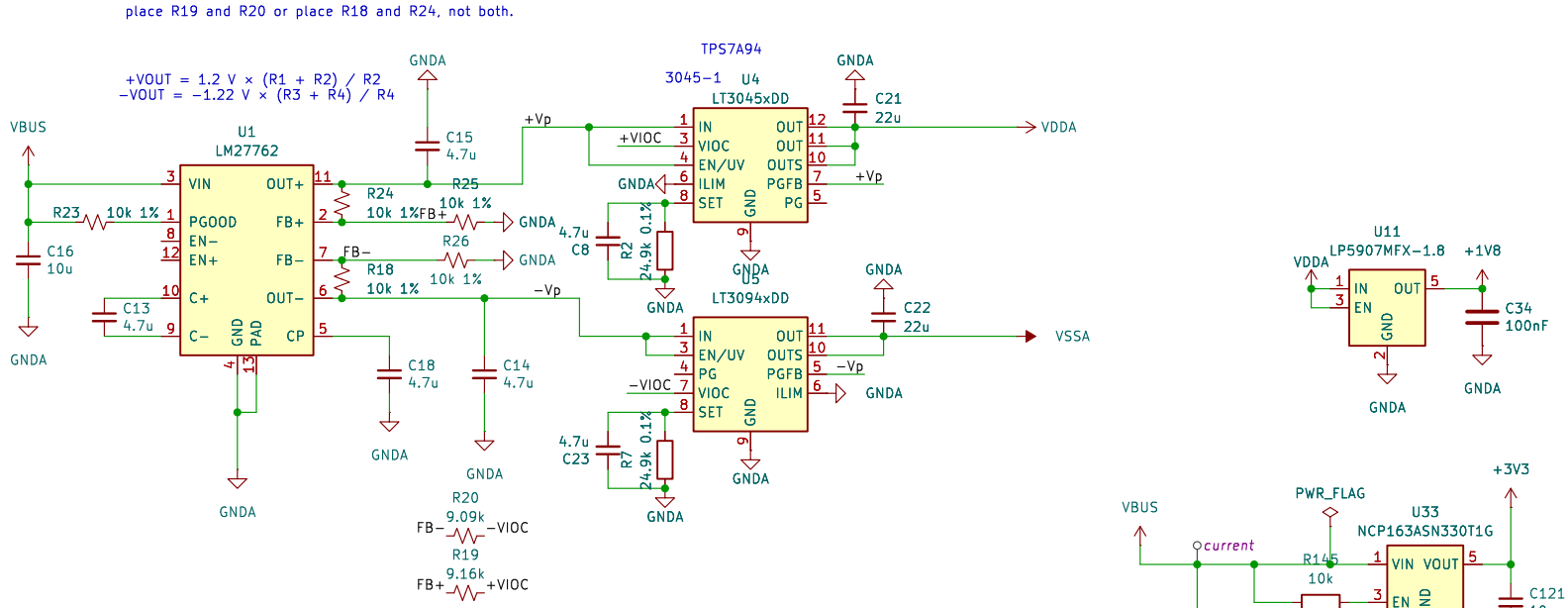


TODO:
update ADC with correct library part and footprint
add voltage reference,

Add precision low noise ADC
Look at Insturmentation amplifiers

Sheet: / File: MR1.kicad_sch		
Title:		
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ADC1:
10 channels: GPIO1 – GPIO10
ADC2 is shared with WiFi
+3V3
NT1
NetTie_2
VCC
TP11
TestPoint
PWR_FLAG
VCC_3V3
C7
10u
C119
10u
R9
10k 1%
C9
4.7u
GND
U8
ESP32-S3-MINI-1U
UART TTL or SPI, selectable in firmware
CHIP_PU
EN
3V3
TXD0
RXD0
TX
RX
R96
R97
27
RXD0
TXD0
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO26
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
GPIO39
GPIO40
GPIO41
GPIO42
GPIO45
GPIO46
GPIO47
GPIO48
USB_D-
USB_D+
LDAC_
CS_
SCK
SDI
DAC_Dout
DAC_CLK
DAC_EN
R33
R36
R10
R37
R39
R40
R28
RTC_crystal_1
RTC_crystal_2
C51
12p
C52
12p
9.0PF
Y2: ESR = Max. 70 KΩ
32.768 kHz
Y2
32.768 kHz
R30
1M
R32
0
R51
0
J22
B4B_PH
J36
Conn_ARM_JTAG_SWD_10
VTref
RESET
SWCLK/TCK
SWDIO/TMS
SWO/TDO
NC/TDI
GNDDetect
JTAG_TCK
JTAG_TMS
JTAG_TDO
JTAG_TDI
SW2
SW_Push
SW1
SW_Push
R34
10k 1%
Jumper
Jumper_1
Boot Mode | GPIO0 | GPIO46
Default Config | 1(pull-up) | 0 (Pull-down)
SPI Boot | 1 | Any Value
Joint DL Boot | 0 | 0
Invalid | 0 | 1
Sheet: /Microcontroller/
File: Microcontroller.kicad_sch
Title:
Size: A
Date:
Rev:
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Sheet: /Power/	
File: Power.kicad_sch	
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Size: USLetter	Date:
KiCad E.D.A. 9.0.6	Rev:
	Id: 3/5

Stage 1: Transimpedance Amp

amplifier board

J10
Conn_01x01_Pin
>PORE_AMP

J11
Conn_01x01_Pin
Vbias_AMP

499Ω Metal Film Resistor

R1 1206

C1 2pF

R1 1M

NT3
NetTie_2
GUARD
Vbias_AMP

J1
Conn_01x01_Pin

H1
Nanopore

J2
Conn_01x01_Pin

R6
1K

Vbias_AMP

VSSA

GUARD

R8
100

Signal_AMP

LMP7721
U2

GUARD

100nF C6

VDDA

GNDA

VDDA VSSA VDDA VSSA

C4 10nF
C5 10nF
C2 100nF
C3 100nF

GNDA GNDA GNDA GNDA

C2 and C3 X5R

P1 >PORE_AMP
GNDA
P2 Vbias_AMP
SHIELD_AMP
P3 Signal_AMP
GNDA
P4 VDDA
VSSA
SHIELD_AMP
SHIELD_AMP MP

J9
Conn_01x10

MG Chemicals 422C conformal coating
PTFE (ZYF255DA): Dk .approx 2.55

low noise voltage converters

Stage 2 Voltage Amplifier

LTC6268

GNDA

1K R5

VDDA

Vbias

Signal_Return

U16
LTC6268xS8

VDDA VSSA

C19 100nF
C20 100nF

GNDA GNDA

R3
1K

R4 100

Vbias

R43 1K
R44 1K

U6A
ADA4940-1xCP

GNDA

R27 27
R41 27

C29 10nF
C30 10nF

GNDA

R45 1K
R46 1K

VDDA VSSA

C31 100nF
C32 100nF

GNDA GNDA

U6B
ADA4940-1xCP

VSSA GNDA

+3V3

+1V8

Vbias

REF VDD VIO
IN+ IN−
GND
SDI SCK SDO CNV

DAC_SDI
DAC_CLK
DAC_SDO
DAC_EN

U10
AD40xxBRMZ

GNDA

+3V3

C33 100nF

GNDA

Vbias
C35 10u

GNDA

P1 >PORE
GNDA
P2
SHIELD
P3 Signal_Return
GNDA
P4 VSSA
VDDA
SHIELD
SHIELD MP

J12
Conn_01x10

NT4
NetTie_2
Vbias SHIELD

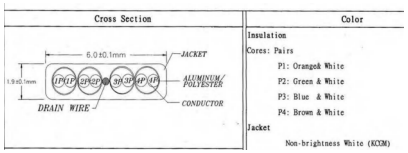
Virtual Ground

consider cat 7 flat ribbon cable, 8 pairs, shielded

pins for holding the board in place while applying adhesive

https://www.mcmaster.com/98378A208/





https://www.vpi.us/bulk-cable-sf/super-flat-cat7-736#idTab9

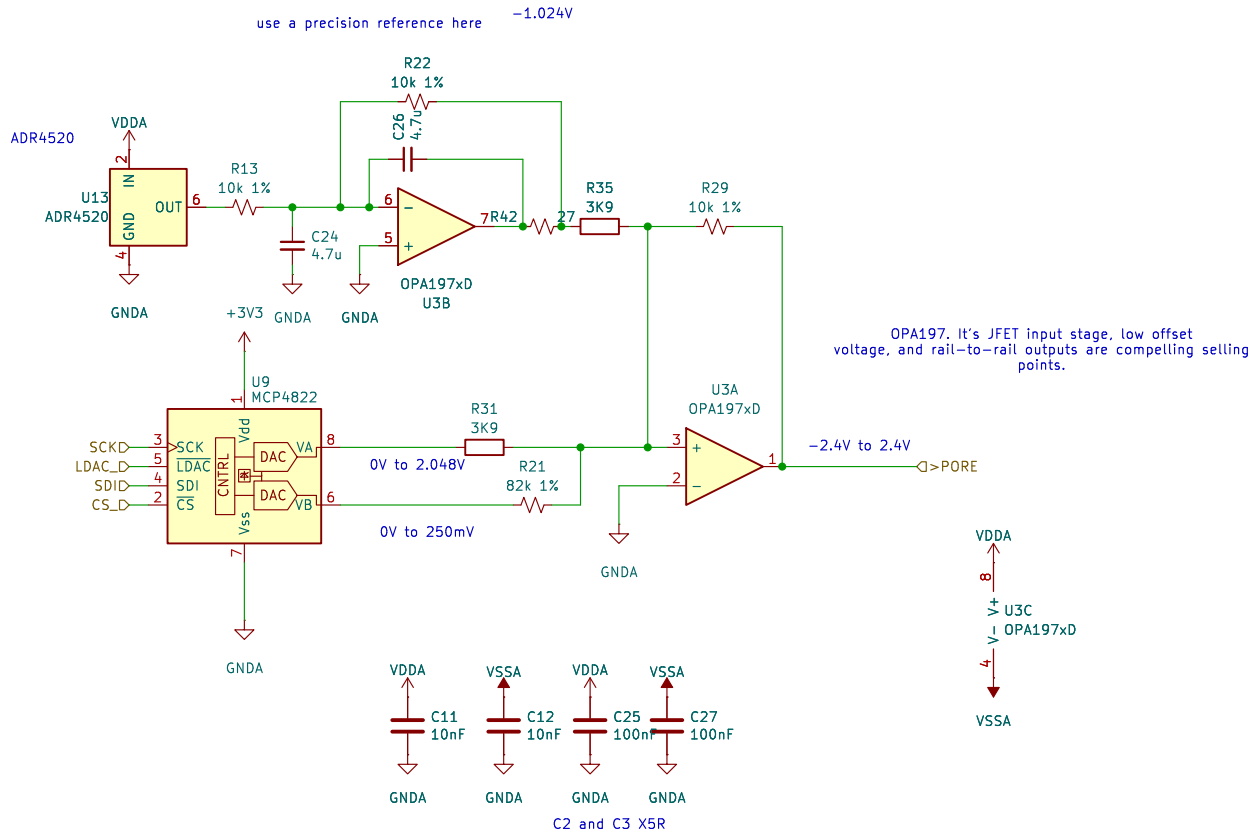
250ft 32AWG CAT7 STP Solid, 0.075" Super Flat Bulk Ethernet Bare Copper Cable – White.

SSH-003T-P0.2-H #32 to #28 (0.032 to 0.081) 0.4 to 0.8 23.000

Sheet: /Insturmentation/
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Sheet: /Ignition/		
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Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 5/5